



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
PRE BOARD EXAMINATION

CLASS: X
SUBJECT: BIOLOGY [SET A]

FULL MARKS: 80
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the Question Paper.
The time given at the head of this Paper is the time allowed for writing the answers.*

*Section A is compulsory. Attempt any four questions from Section B.
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of Nine printed pages.*

SECTION A

(Attempt all questions from this Section)

Question 1

Choose the correct answers to the questions from the given options. (Do not copy the question. Write the correct answer only):

[15]

- (i) A 55-year-old woman possesses some symptoms of Exophthalmic goitre. Identify the symptom/s of it:

P- Marked increase in metabolic rate.

Q- Eyes are protruded.

(a) Only P

(b) Both P and Q

(c) Only Q

(d) Neither P nor Q

- (ii) On which day of the menstrual cycle does ovulation take place?

(a) 1st day

(b) 5th day

(c) 14th day

(d) 21st day

- (iii) David tries to understand an important concept of plant physiology. He has experimented on a marine fish and found that when it was placed in tap water it bursts. This is because of:

1. P- Endosmosis

2. Q- Plasmolysis and Exosmosis

3. R- Exosmosis and Active Transport

4. S- Active transport

(a) Only P

(b) Both P and Q

(c) Only R

(d) Both R and S

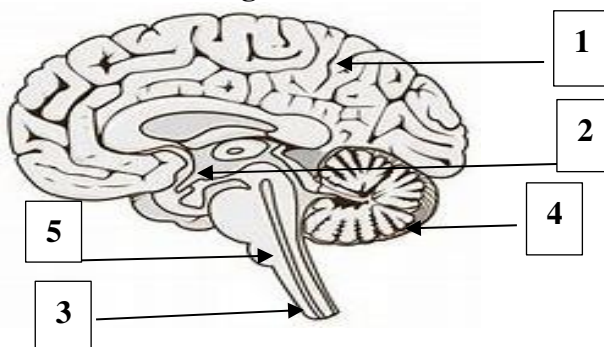
- (iv) **Assertion (A):** Roots are the primary organs responsible for the absorption of water in plants.
Reason (R): The root hairs present on the surface of the roots significantly increase the surface area, allowing for efficient absorption of water and minerals from the soil.
- (a) Both A and R are false (b) A is true and R is false
(c) A is false and R is true (d) Both A and R are true
- (v) Ananya was preparing her Biology lessons on human eye. Four of her friends tried to help Ananya understand the sequence of events that take place when human eye changes its focus from a distant object to an object closer to the eye.
Identify who had given the correct answer:
- (a) Neha- ciliary muscles relax → curvature of lens increases → focal length of eye increases
(b) Jaya- ciliary muscles contract → curvature of lens decreases → focal length of eye increases
(c) Sushma- ciliary muscles contract → curvature of lens decreases → focal length of eye decreases
(d) Praveen- ciliary muscles contract → curvature of lens increase → focal length of eye decreases
- (vi) Mrs. Pal is a mother of two kids. She and her husband thought of family planning and they decided to go for an operative method to control child birth. Identify the process which Mr. Pal will go through from the following:
- (a) Ovectomy (b) Vasectomy
(c) Tubectomy (d) Histotomy
- (vii) Dr. Srivastava was recording the pulse rate of Suman. Where exactly does he press on Suman's wrist?
- (a) Capillary (b) Nerve
(c) Vein (d) Artery
- (viii) **Assertion (A):** In human male, testes are extra abdominal organs which are present inside scrotum.
Reason (R): Scrotum has a relatively lower temperature needed for the production and storage of sperms.
- (a) Both A and R are true (b) A is true and R is false
(c) A is false and R is true (d) Both A and R are false
- (ix) The *Homo erectus* was the earliest prehistoric hunter-gatherers who were highly successful in adapting to different habitats across the Old World.
The cranial capacity of *Homo erectus* was about:
- (a) 650 c.c. (b) 900 c.c.
(c) 1200 c.c. (d) 1600 c.c.

- (x) Harry was travelling to the Thar Desert with his family on a vacation. He observed the leaves of certain plants being modified into spines. The diagram of such a plant is given below:



This was an adaptation of the plant to cope with:

- | | |
|-------------------------|----------------------------|
| (a) Excess humidity | (b) Reducing transpiration |
| (c) Excessive guttation | (d) Reducing absorption |
- (xi) Prerna was drying clothes on the terrace. Suddenly a monkey came near her. She ran with full speed. Which of the following are involved in the process mentioned above?
- | | |
|---------------|-------------------|
| 1. Adrenaline | 2. Nor-adrenaline |
| 3. Insulin | 4. Cortisones |
- | | |
|------------------|---------------------|
| (a) Only 3 | (b) Only 1 |
| (c) Only 1 and 2 | (d) Only 1, 3 and 4 |
- (xii) Two pea plants, one with round, green seeds (RRYY) and another with wrinkled, yellow (rryy) seeds, produce F₁ progeny that have round green (RrYy) seeds. When F₁ plants are self-pollinated, the F₂ progeny will have a new combination of characters. Choose the new combinations which are observed in F₂ generation from the following:
- | | |
|---------------------|--------------------|
| 1. Round, yellow | 2. Round, green |
| 3. Wrinkled, yellow | 4. Wrinkled, green |
- | | |
|-------------|-------------|
| (a) 1 and 2 | (b) 1 and 3 |
| (c) 2 and 3 | (d) 1 and 4 |
- (xiii) A little confused with the internal structure of human brain Rashmi needs help from you to identify the labelled parts. Help her in identifying the parts labelled from 1-5 in the diagram below:



- | |
|--|
| (a) 1-Cerebrum, 2- Hypothalamus, 3- Spinal cord, 4- Cerebellum, 5- Medulla |
| (b) 1-Cerebrum, 2- Pituitary gland, 3- Spinal cord, 4- Cerebellum, 5- Pons |
| (c) 1-Cranium, 2- Hypothalamus, 3- Medulla, 4- Cerebrum, 5- Pons |
| (d) 1-Cerebellum, 2- Hypothalamus, 3- Spinal cord, 4- Cerebrum, 5- Medulla |

(xiv) Assertion (A): Fertilization of the ovum and sperm occurs in the sperm duct of the female reproductive system.

Reason (R): Vas deferens carries sperms from the epididymis to the urethra of the male reproductive system.

- (a) Both A and R are false (b) A is true and R is false
(c) A is false and R is true (d) Both A and R are true

(xv) The hormone which maintains water balance of our body:

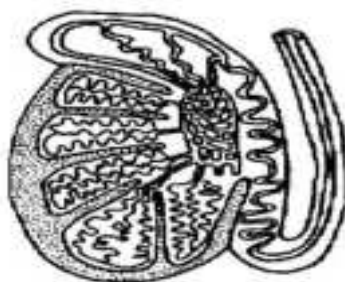
- (a) TSH (b) ACTH
(c) LH (d) ADH

Question 2

(i) Name the following: [5]

- (a) The full form of ACTH.
(b) Part of the brain which controls body temperature.
(c) The part of the ear which converts air vibration into fluid vibration.
(d) The most light sensitive region of the retina.
(e) Onset of menstrual cycle in a human female.

(ii) The diagram given below is the longitudinal section of human testis. Read the information given below the diagram and fill in the blanks (a) to (e): [5]



Human males have two testicles of similar size contained within the __ (a) __ which is an extension of the abdominal wall. The testis contains very fine coiled tubes called __ (b) __, which produces __ (c) __ and interstitial cells are responsible for the production of the sex hormone __ (d) __. The site of maturation of sperm is known as __ (e) __.

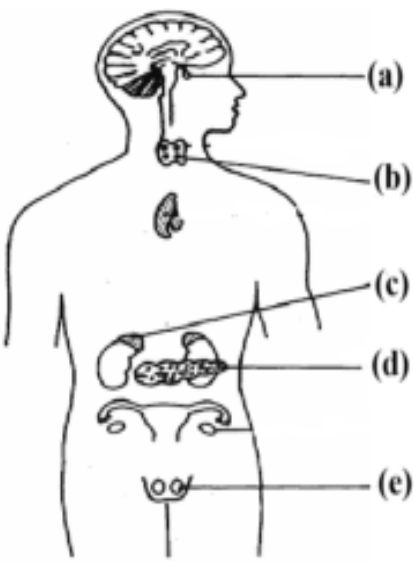
(iii) Arrange and rewrite the terms in each group in the correct order so as to be in a logical sequence beginning with the term that is underlined. [5]

- (a) Water molecules, Grana, Photons, Hydrogen and Hydroxyl.
(b) Afferent arteriole, Renal Artery, Efferent arteriole, Glomerulus.
(c) Neanderthal man, Homo erectus, Cro-Magnon, *Homo sapiens*.
(d) S, G₂, M, G₁.
(e) Stoma, Mesophyll cells, Intercellular Space, Sub stomatal space.

(iv) Read the explanation given below and name the structure: [5]

- (a) Phase which is considered to be the reversal of Prophase.

- (b) The ability to perceive depth or relative distance of the objects due to the simultaneous focusing of both eyes.
- (c) Part of the female reproductive system in which the zygote gets implanted.
- (d) A radiation pollutant which caused the risk of thyroid cancer released from Fukushima Daiichi Nuclear Disaster.
- (e) Fluid part of blood.
- (v) Given below is a human torso to show the location of different Endocrine glands of our body. Match the structures marked (a) to (e) with their correct names or functions: [5]

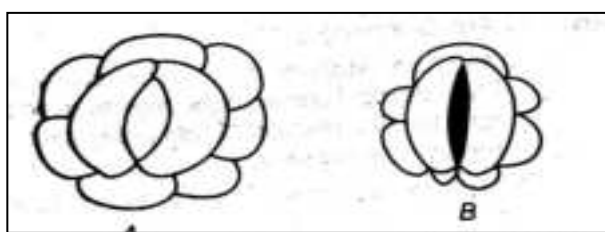
The Endocrine glands of our body	Names/Functions
	1. Controls basal metabolism. 2. Releases Tropic hormones. 3. A myxocrine gland 4. Releases relaxin to control labour pain 5. Also known as supra-renal gland 6. Releases the hormone testosterone

SECTION B (40 Marks)

(Attempt any four questions from this section)

Question 3

- (i) Define Transpiration. [1]
- (ii) Religious Customs and Old beliefs are the major causes of population explosion. Explain briefly. [2]
- (iii) An oil spill is the release of a liquid petroleum hydrocarbon into the marine ecosystem, due to human activity, and is a form of pollution. Describe the harmful effects of Oil Spill. [2]
- (iv) Observe the diagram and answer the questions that follow: [2]



- (a) Which guard cells show high water content A or B?
- (b) State the condition of the guard cells A and B.

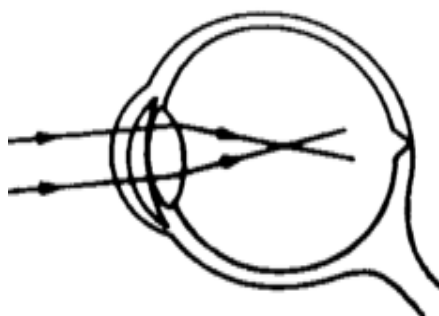
- (v) Draw a neat labelled diagram to bring out the differences between an artery and a vein. [3]

Question 4

- (i) Name any two nitrogenous bases present in DNA. [1]
(ii) Distinguish between the following pairs- Osmotic pressure and Turgor Pressure on the basis of their definition. [2]
(iii) Aryan was watching the display of fireworks in the sky. [2]



- (a) Trace the path of the light rays using the following terms:
Fovea, Lens, Conjunctiva, Pupil, Cornea.
(b) Name the nerve that carries the impulse for vision to the brain.
(iv) A geneticist wishes to design a cross between a colour blind male and a normal female. What percentage of normal sons can be expected from such a cross? Support your answer with a Punnett Square. [2]
(v) Rishav visits an ophthalmologist for his regular checkup. His doctor shows him this figure explaining an eye defect which he is likely to get. [3]



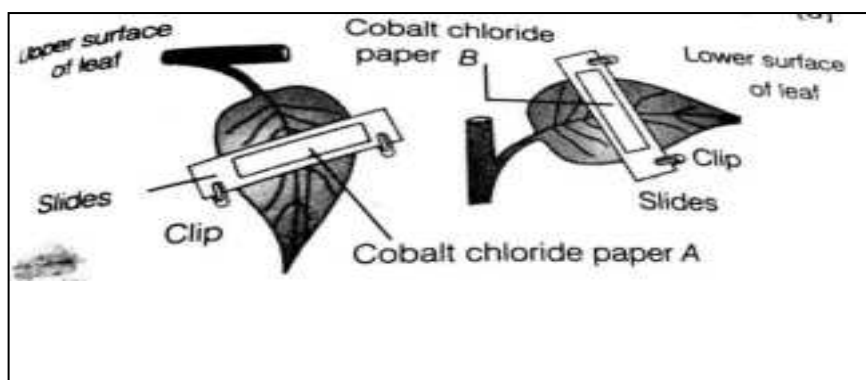
- (a) Give the scientific term for this defect.
(b) State one possible reason for the defect.
(c) With the help of a labelled diagram show how this eye defect can be rectified.

Question 5

- (i) Which blood vessel carries the highest amount of urea in mammals? [1]
(ii) Nuclear pollution or radiation pollution marks as the release of high energy particles into the air, water, or surface by the accumulation of radioactive substances in the environment, posing serious health risks. Mention two harmful effects of radioactive pollution. [2]

(iii) Study the diagram given below and answer the questions that follow:

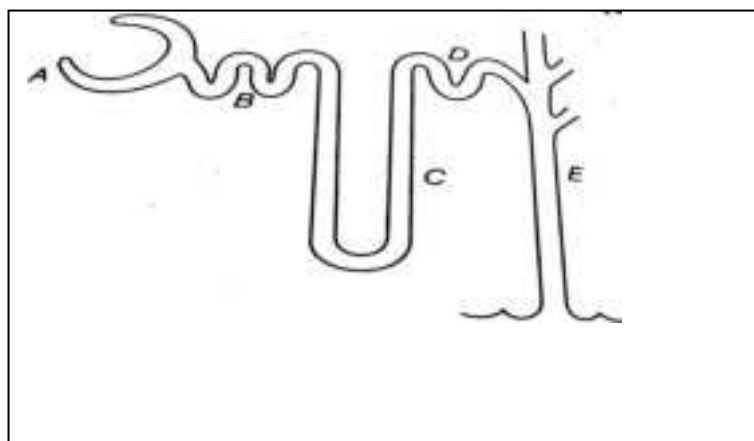
[2]



- (a) What is the aim of the experiment?
- (b) What conclusion can we draw from this experiment based on the cobalt chloride papers A and B?
- (iv) Differentiate between the following pairs based on the criteria given in brackets:
Homozygous and Heterozygous allele (definition).
- (v) The diagram given below represents basic structure of a renal tubule.

[2]

[3]



- (a) Name the part labelled as B.
- (b) Mention the function of part labelled C.
- (c) Briefly explain the function of part labelled A.

Question 6

- (i) Explain the term-Monohybrid Cross.
- (ii) Natural selection enhances the preservation of a group of organisms that are best adjusted to the physical and biological conditions of their environment and may also result in their improvement in some cases. Define- Natural Selection. Which evolutionist suggested this theory?
- (iii) Expand the abbreviation FSH. State its main function in female reproduction.
- (iv) Name the cells that secrete insulin. What will be its effect on sugar level in blood, if the secretion is insufficient?

[1]

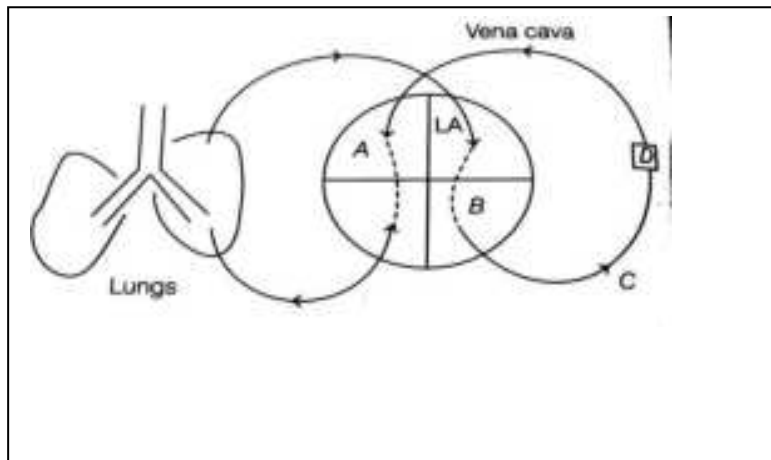
[2]

[2]

[2]

- (v) Medha peeped into the medical books of her elder sister and found out this diagram. Help her understand the circulatory system in man better by answering the questions that follow:

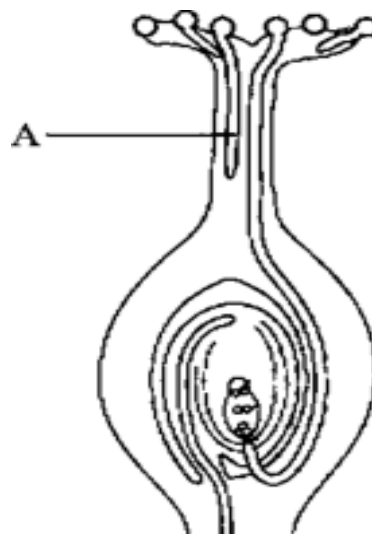
[3]



- Identify the parts labelled as B in the above diagram.
- Which valve is found at the beginning of the part labelled as C?
- Mention the function of the part labelled as C.

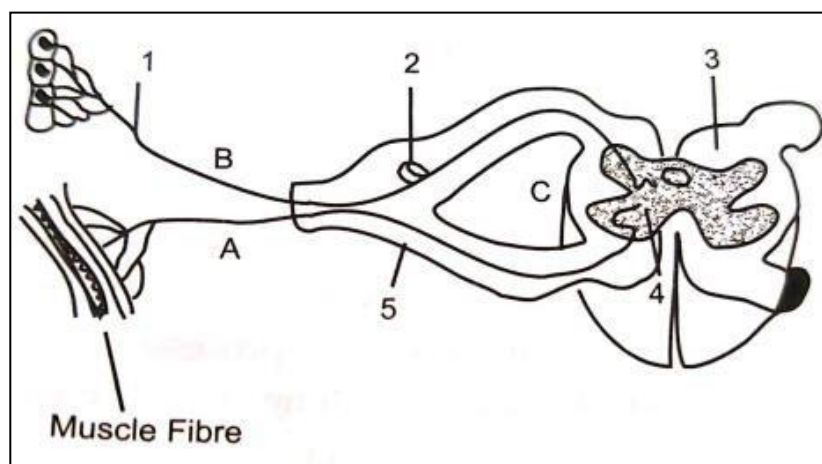
Question 7

- Name the instrument which helps in measuring blood pressure. [1]
- Identify the plant movement exhibited by part labelled as 'A'. Define it. [2]



- The sympathetic and parasympathetic nervous systems are two parts of the autonomic nervous system (ANS) that work together to keep the body in balance. Differentiate between parasympathetic and sympathetic nervous system on the basis of their location and function. [2]
- Mention the functions of Seminal vesicle and Bulbo-Urethral gland in the male reproductive system. [2]

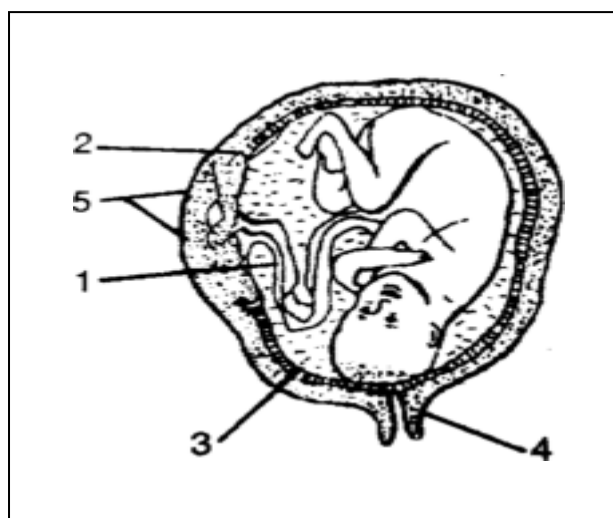
- (v) The diagram given below shows the internal structure of the spinal cord, depicting a simple reflex. Study the same and answer the questions that follow: [3]



- Label the parts 3.
- Using the letters of the alphabet shown in the figure, indicate the direction in which an impulse enters and leaves the spinal cord.
- What is the term given to the point of contact between two nerve cells?

Question 8

- Expand the abbreviation NADP. [1]
- Mention any two objectives of Swachh Bharat Abhiyan. [2]
- A middle-aged person recently developed an eye defect which is known as Presbyopia. Define Presbyopia and mention the lens used to correct this eye defect. [2]
- Mention the two functions of Aqueous Humour. [2]
- During her third trimester of pregnancy, Priya was suggested ultrasonography by her gynaecologist. She was overwhelmed to see such a condition of her womb. The diagram below is that of a developing human foetus in her womb. [3]



- What is the term given to the period of development of the foetus in the womb?
- Name the hormone produced by the part labelled '2'.
- State the function of the part labelled as "1".

